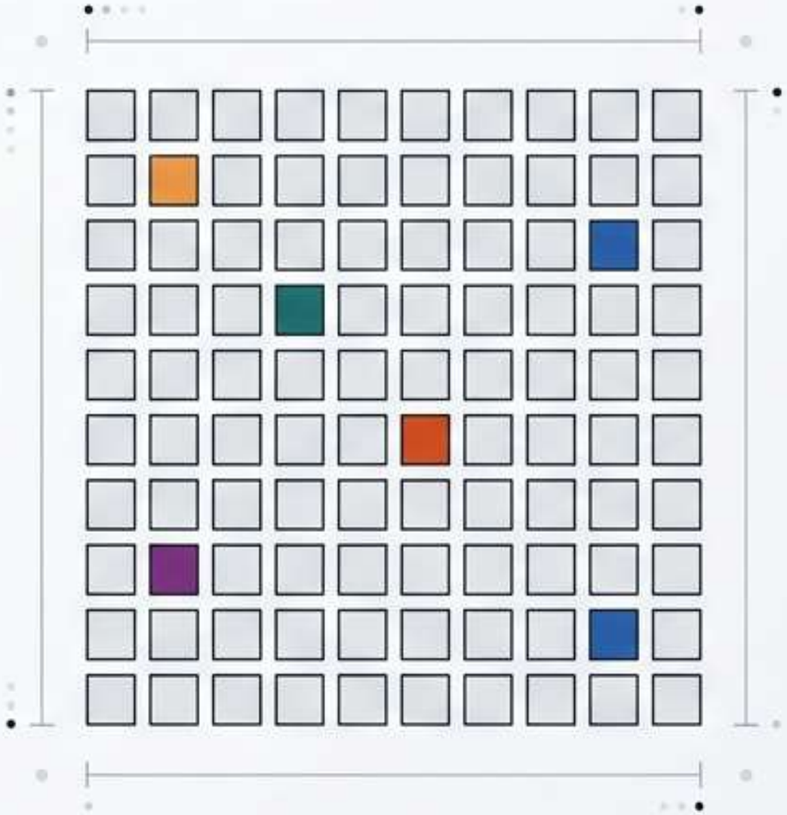
Signal in the Noise: Supervised Classification of Pediatric Brain Cancer via High-Dimensional Transcriptomics

```
dataset_source = 'CuMiDa Affymetrix U133 Plus 2.0'
n_samples = 130
n_features = 54676
```

PREPARED FOR REVIEW: MONASH UNIVERSITY AI & BIOMEDICINE RESEARCH

The Molecular Signature

Brain cancer is characterized by distinct cellular genetic alterations. Whole genome sequencing and microarray data capture these large-scale transcriptomic shifts, mapping biological disease states to raw gene expression numbers.

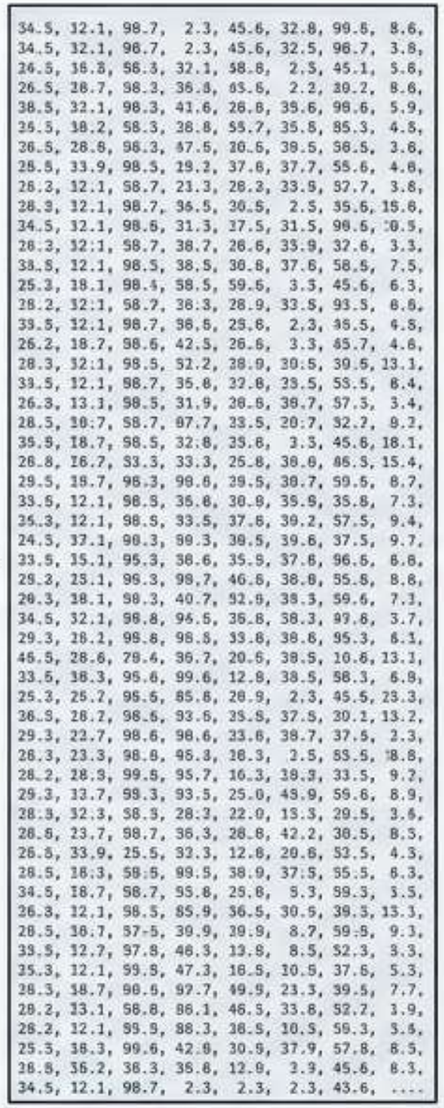


The Algorithmic Target

To train, tune, and evaluate supervised machine learning models capable of accurately differentiating between five specific disease states using unreduced, post-processing expression data.

- **Normal (0)**
- **Ependymoma (1)**
- **Glioblastoma (2)**
- **Pilocytic Astrocytoma (3)**
- **Medulloblastoma (4)**

54,676 Features
(Gene Expressions)



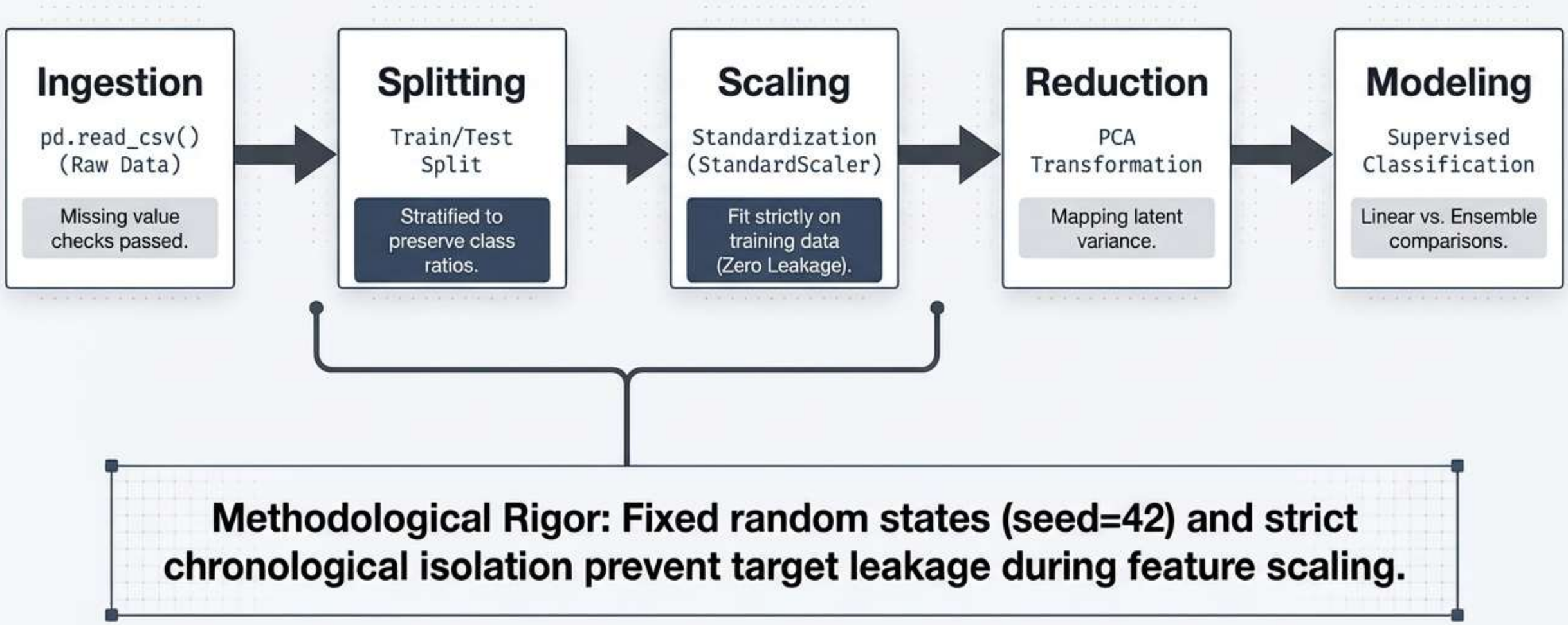
130 Samples

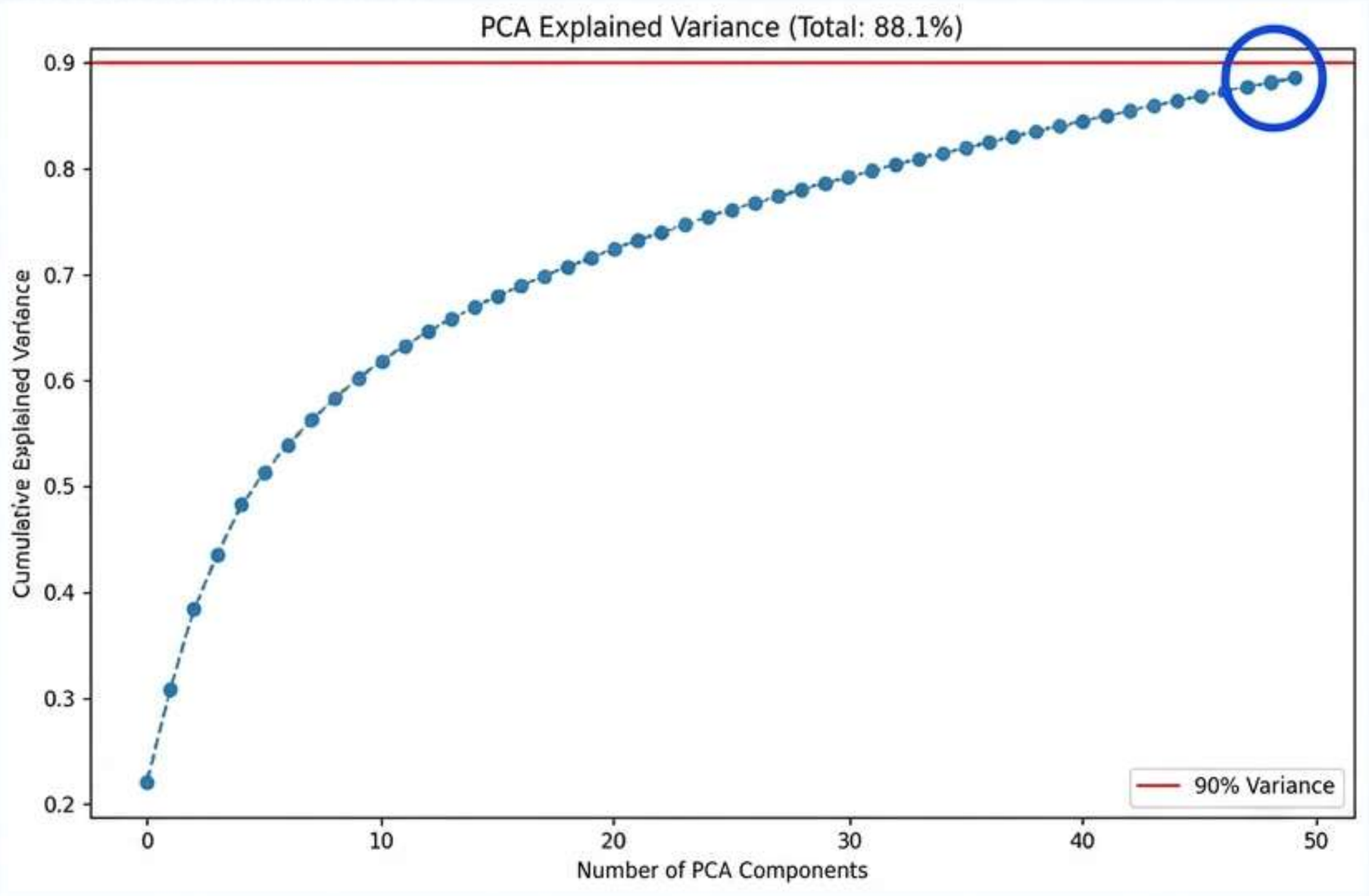
This $p \gg n$ paradigm defines the "Curse of Dimensionality." Training models directly on this matrix risks severe overfitting.

Class Imbalance

```
count(Ependymoma) == 34
count(Normal) == 10
...
```

Key Takeaway: Robust dimensionality reduction and weighted evaluation metrics are strict mathematical necessities, not optional enhancements.


Ingestion
pd.read_csv()
(Raw Data)
Missing value checks passed.
Splitting
Train/Test Split
Stratified to preserve class ratios.
Scaling
Standardization
(StandardScaler)
Fit strictly on training data (Zero Leakage).
Reduction
PCA Transformation
Mapping latent variance.
Modeling
Supervised Classification
Linear vs. Ensemble comparisons.
Methodological Rigor: Fixed random states (seed=42) and strict chronological isolation prevent target leakage during feature scaling.



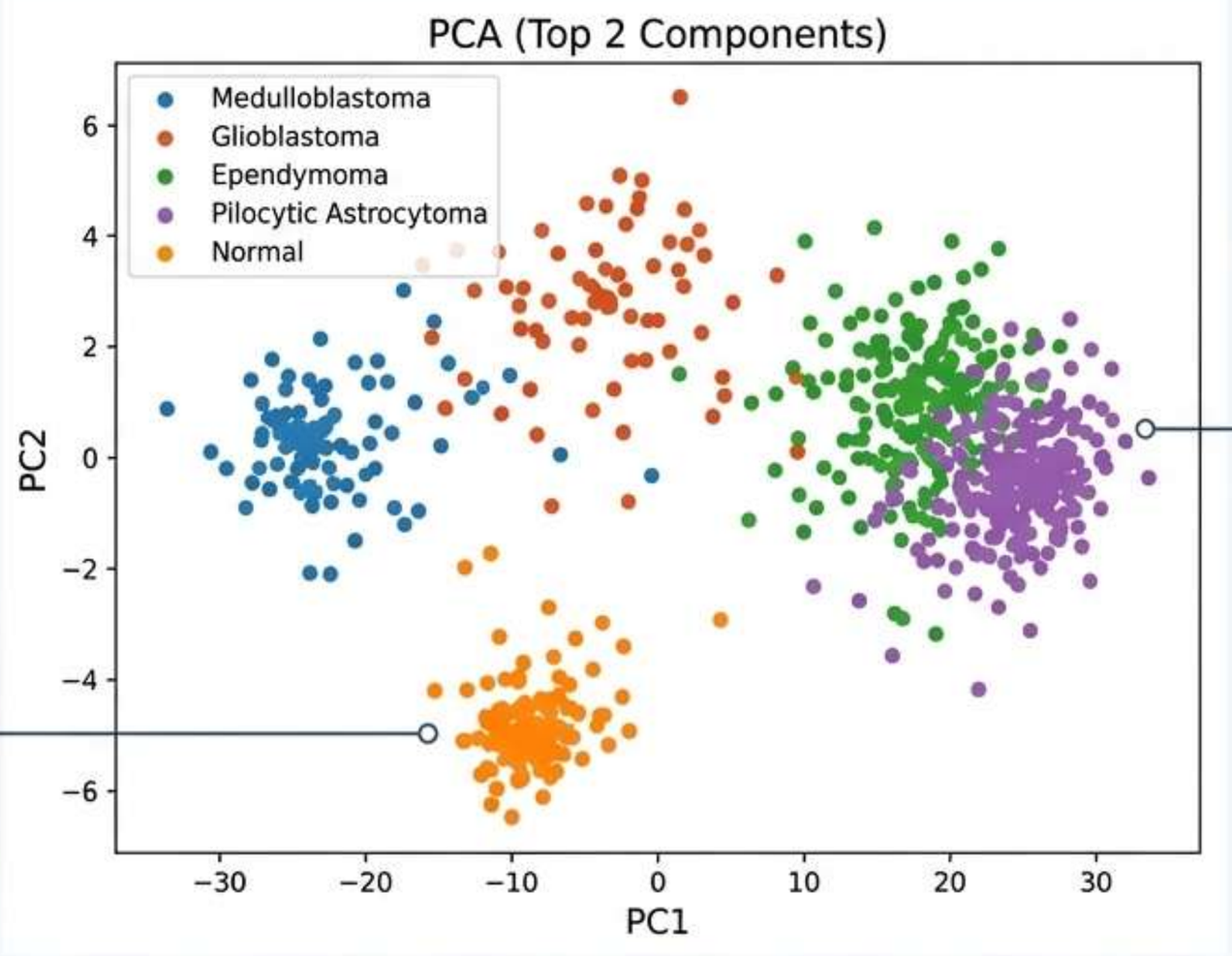
Mathematical Compression

The Principal Component Analysis (PCA) successfully distills the 54,676-dimensional space.

Reaching the 90% cumulative explained variance threshold requires fewer than 50 components. This confirms that the vast majority of transcriptomic features are either collinear or represent biological background noise.

```
dim_reduction_ratio ≈ 1000:1
```

PCA (Top 2 Components) - Training Data



1

Distinct Biological State: Normal samples isolate cleanly in the primary component space.

2

Pathological Proximity: Ependymoma and Pilocytic Astrocytoma share closer latent geometries, predicting a higher baseline for model confusion.

Key Takeaway: The 2D projection confirms that distinct cancer types express separable molecular signatures, validating the transition to predictive modeling.

	Logistic Regression	Decision Trees	Ensemble Methods (RF/XGBoost)
Model Architecture	Linear Mapping	Hierarchical Splits	Non-linear Ensembles
Theoretical Fit for Genomics	Strong baseline interpretability.	Prone to overfitting on small samples.	Theoretically capture complex, non-linear gene expression interactions.
Handling High Dimensionality	Efficient in compressed PCA space.	Struggles with collinearity.	Robust against noise, handles varying feature importance.

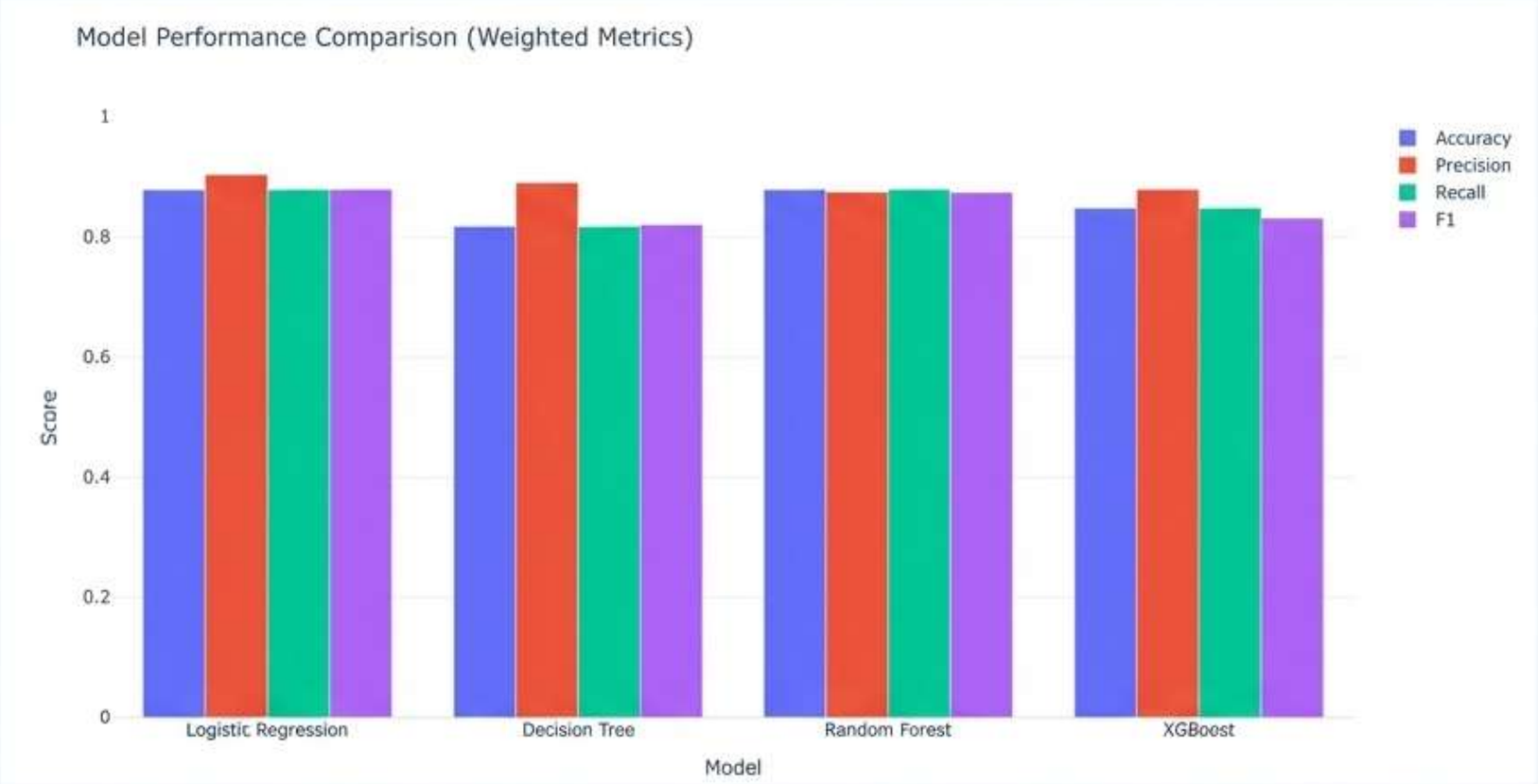
Evaluation Protocol

Due to the severe class imbalance, standard accuracy is discarded in favor of Weighted Precision, Recall, and F1-Scores to prevent majority-class bias.

Peak_Accuracy:
~90-95%

Best_Linear:
Logistic Regression

Best_Ensemble:
Random Forest



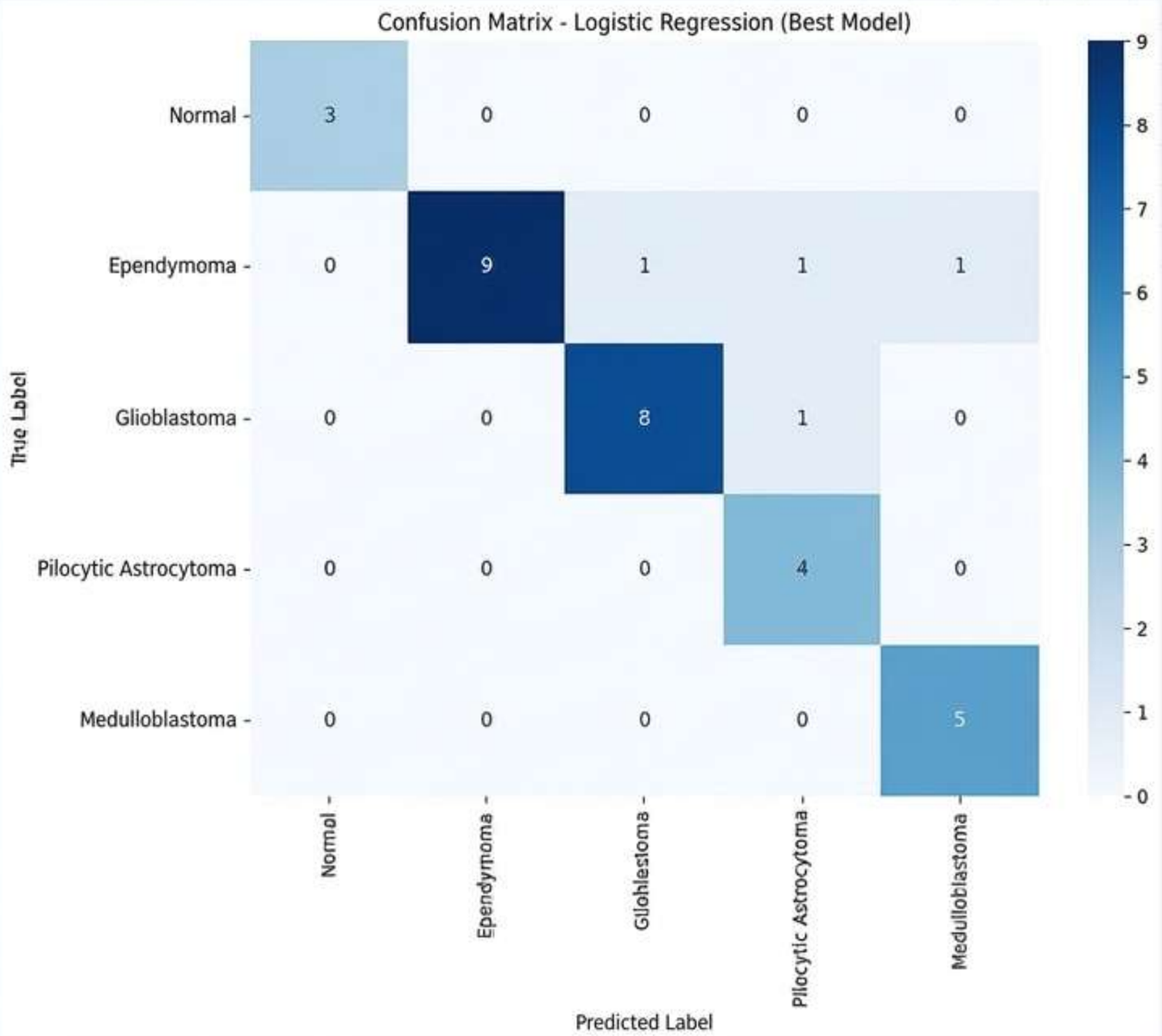
While tree-based ensembles (XGBoost/RF) are theoretically superior for high-dimensional non-linear mapping, the PCA-compressed genomic data proved highly linearly separable. Consequently, Logistic Regression performed on par with Random Forest, offering high efficiency with lower computational overhead.

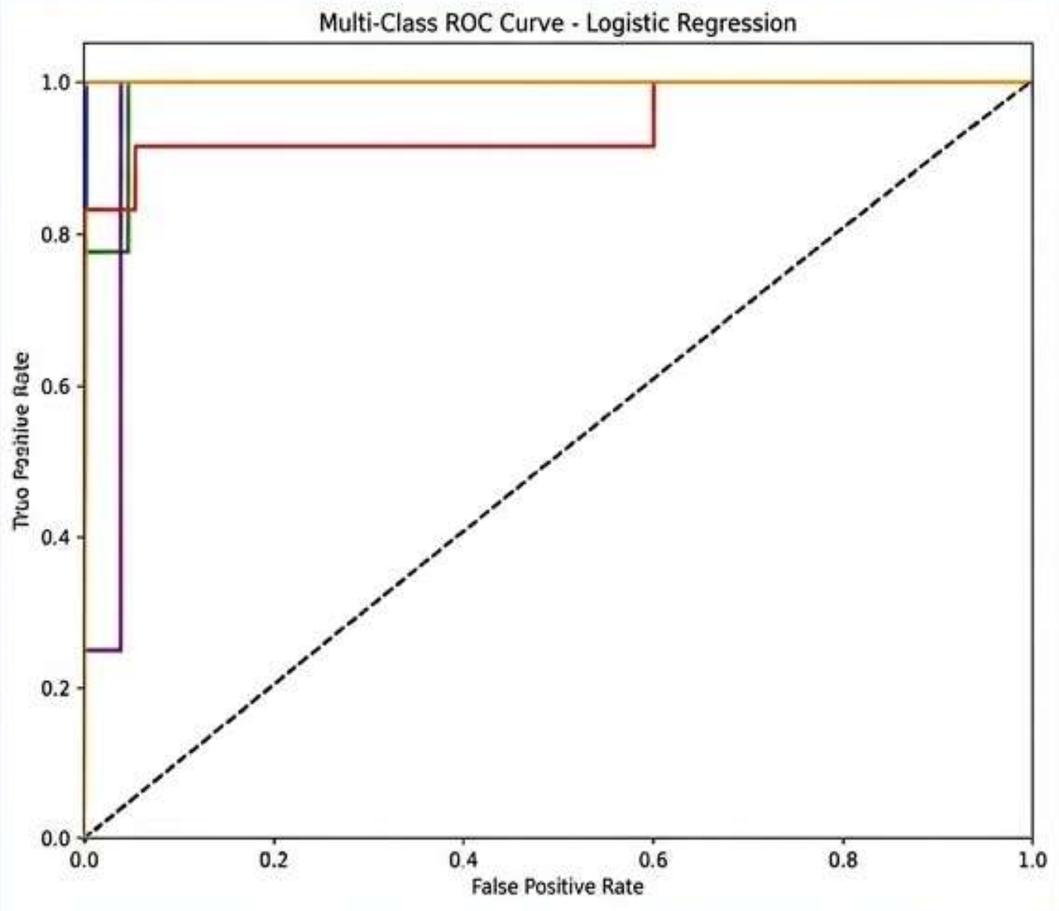
Granular Misclassifications

Perfect Isolation: Normal (3/3) and Medulloblastoma (5/5) achieved perfect classification in this test split, corroborating their **distinct latent clustering**.

The Overlap: The model misclassified 1 true Glioblastoma as Pilocytic Astrocytoma.

Biological Context: Rather than a mere mathematical penalty, this specific confusion highlights the **shared transcriptomic pathways** among distinct glial-derived tumors.

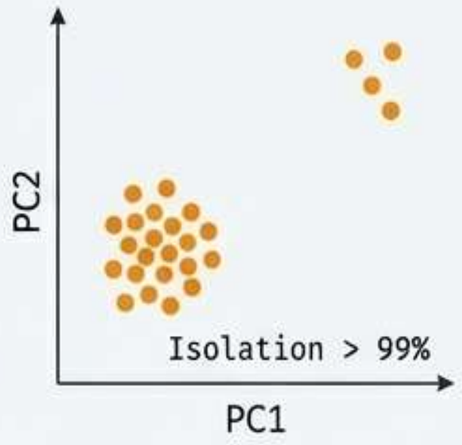


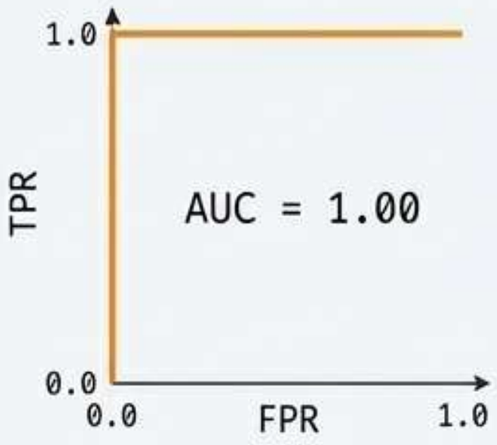


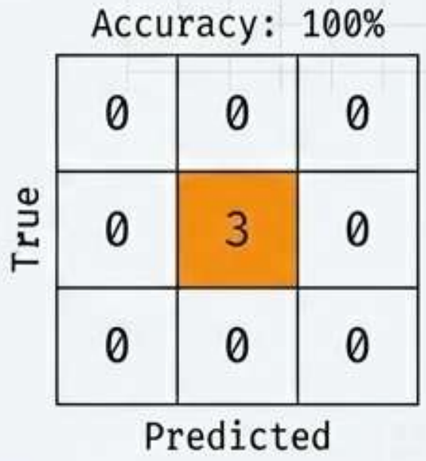
- Normal: AUC = **1.00**
- Medulloblastoma: AUC = **1.00**
- Glioblastoma: AUC = **0.99**
- Pilocytic Astrocytoma: AUC = **0.97**
- Ependymoma: AUC = **0.94**

Concluding Insight Bo: The near-perfect Area Under the Curve across all subtypes indicates that despite the high-dimensional noise of the raw array, the extracted molecular signal is exceptionally robust across varying classification thresholds.

Geometric Isolation → Predictive Confidence → Diagnostic Accuracy







"The topology of the latent space directly dictates predictive efficacy. Where **gene expression** signatures overlap geometry, classification boundaries **blur probabilistically. Algorithm selection** must adapt to this underlying biological geometry."

Computational Stack

Production-ready **`Python`**. Advanced **`Pandas`** manipulation. **`Scikit-learn pipelining`** (PCA, LogReg, Ensembles). **`Plotly/Seaborn`** data visualization.

Methodological Rigor

Strict data hygiene. Pre-split stratification. Complete elimination of data leakage during scaling and reduction. Fixed random states for total reproducibility.

High-Dimensional Analytics

Navigating the $p \gg n$ problem. **Dimensionality reduction**. Applying weighted metrics for severe class imbalance. Translating mathematical output to biological relevance.

Established a robust, reproducible preprocessing and classification baseline for pediatric brain cancer transcriptomics, achieving ~90-95% accuracy via signal-optimized dimensionality reduction.

Ready for Scale

This architectural framework is modular. The next step is applying these validated pipelines to larger, multi-omics datasets (integrating methylation or clinical covariate data) within an active AI & Biomedicine research lab.